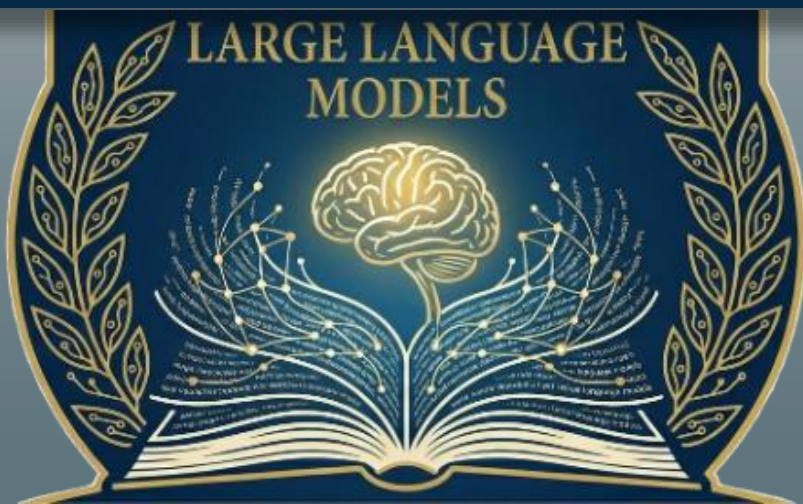


Scaling Laws for LLMs



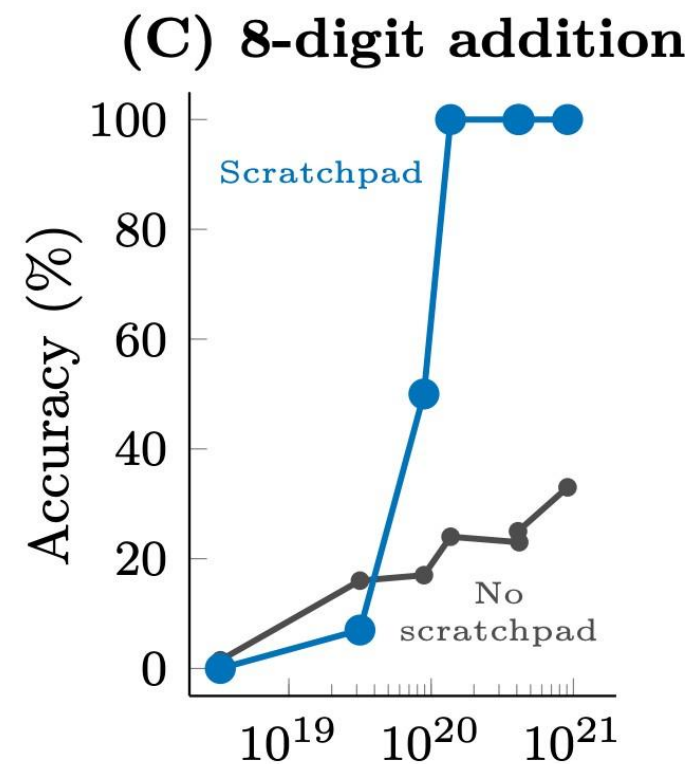
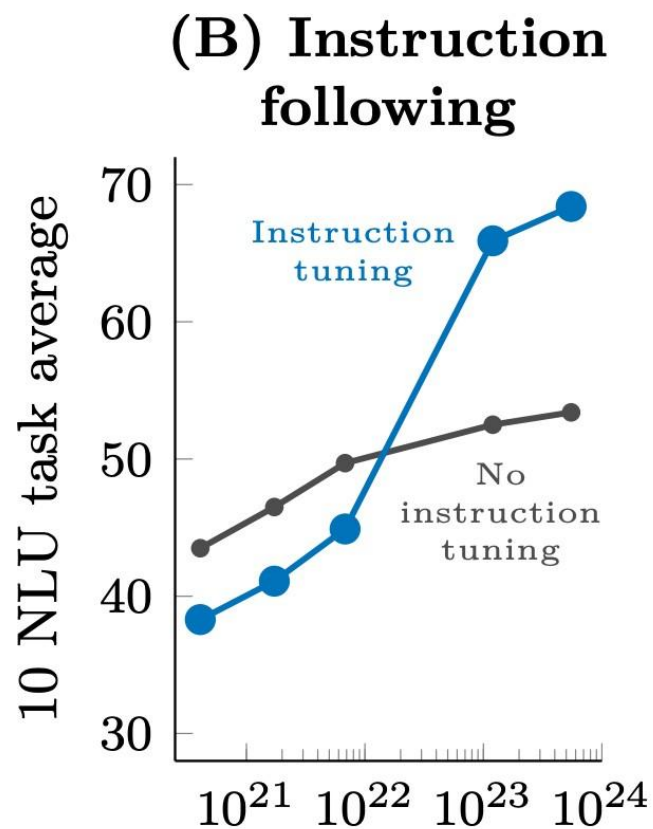
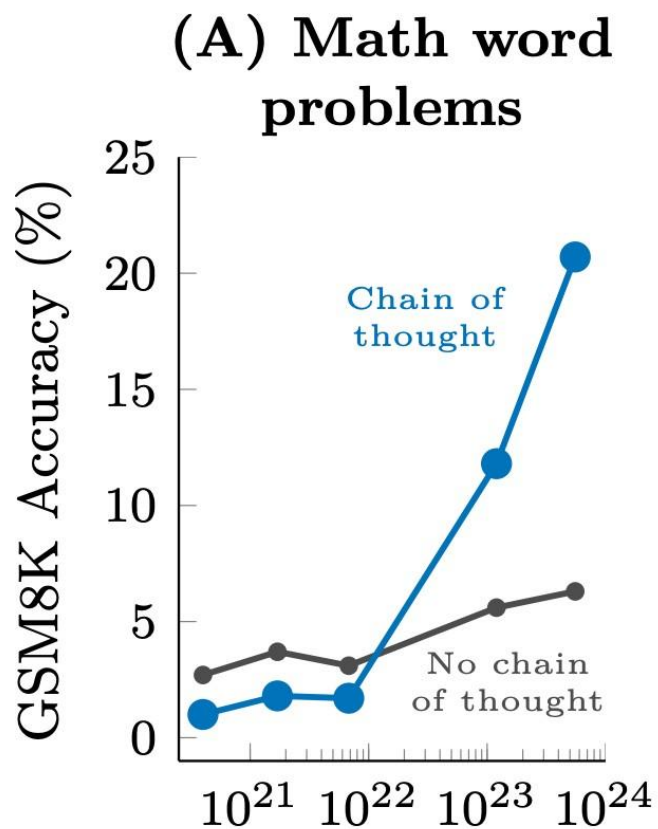
Mohammad Javad Dousti

Sources:

- Mohit Iyyer's Advanced NLP's slides from U of Maryland.

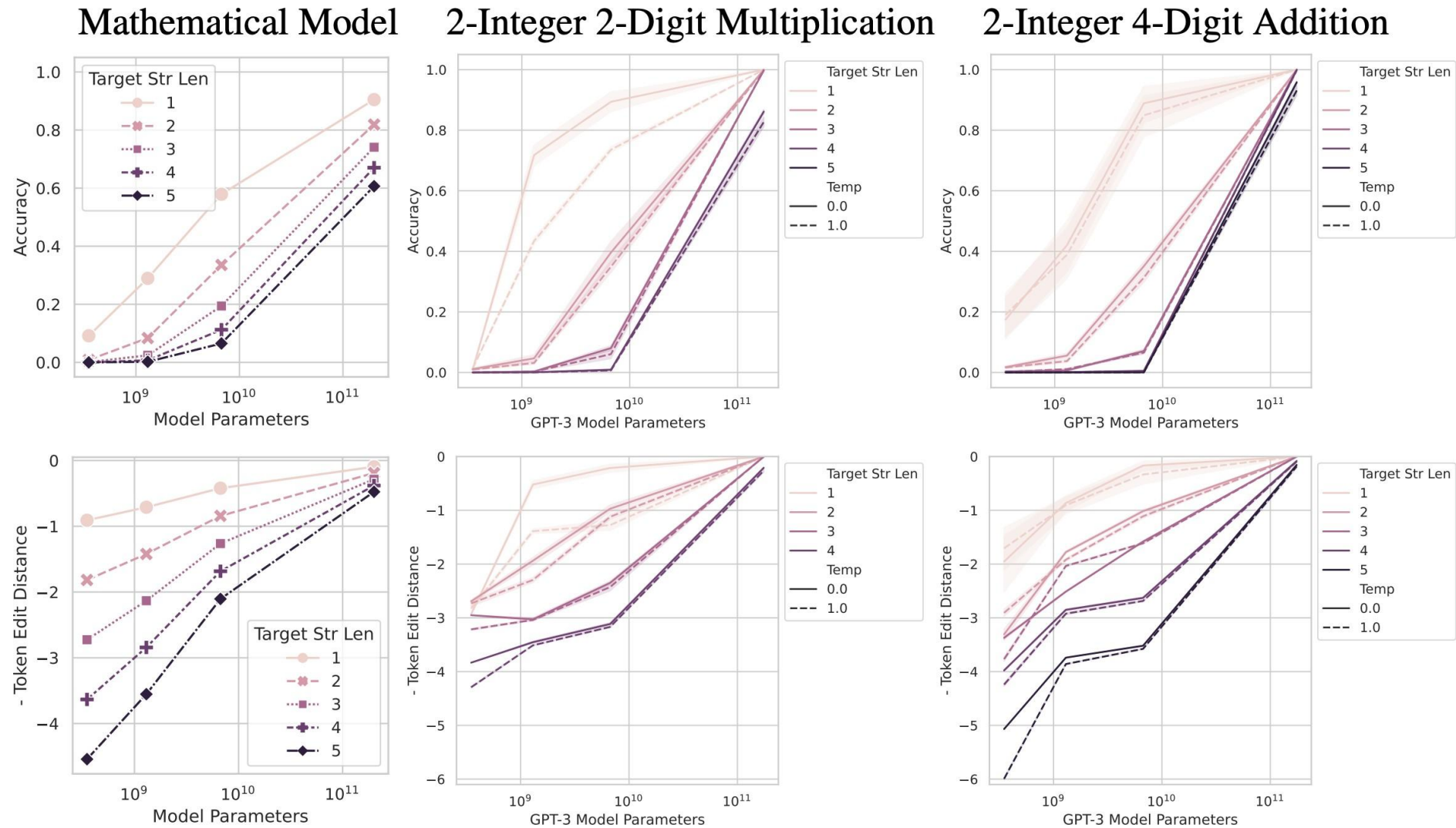
What do bigger LMs buy us?

- ❑ “In-context” learning, chain-of-thought prompting, instruction following, more memorized knowledge and patterns from the training data, etc.
- ❑ Broadly, “*emergent properties*”, which may only appear with larger LMs but not smaller ones
- ❑ *“The ability to perform a task via few-shot prompting is emergent when a model has random performance until a certain scale, after which performance increases to well-above random.”* (Wei et al., 2022)



Model scale (training FLOPs)

Are “*emergent properties*” really emergent?



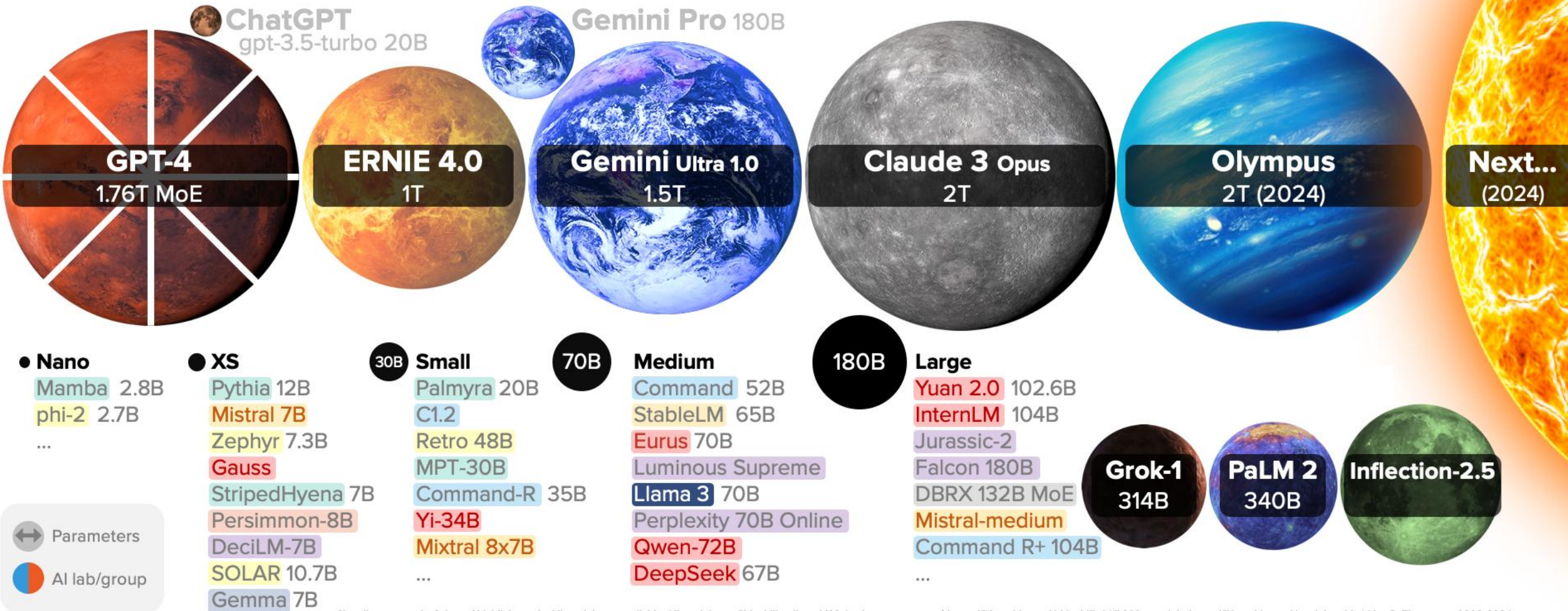
Source: Schaeffer et al. “Are Emergent Abilities of Large Language Models a Mirage?”, NeurIPS 2023

What can we scale?

- ❑ Model size
- ❑ Dataset size
- ❑ Amount of total compute used during training (e.g., number of training steps)



LARGE LANGUAGE MODEL HIGHLIGHTS (APR/2024)

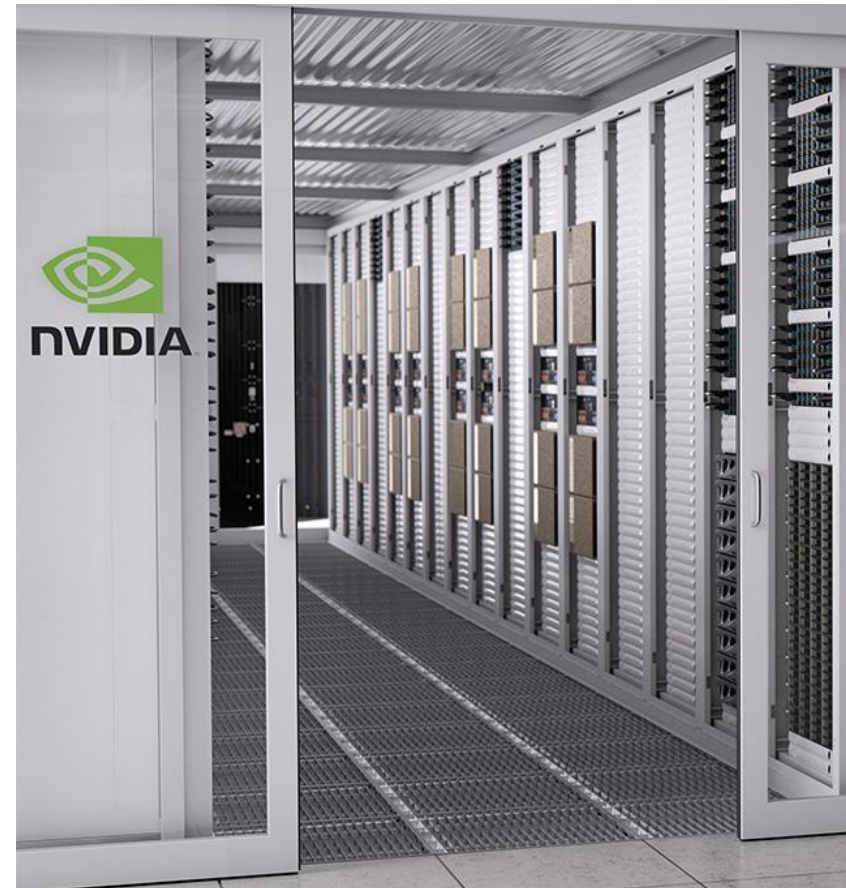
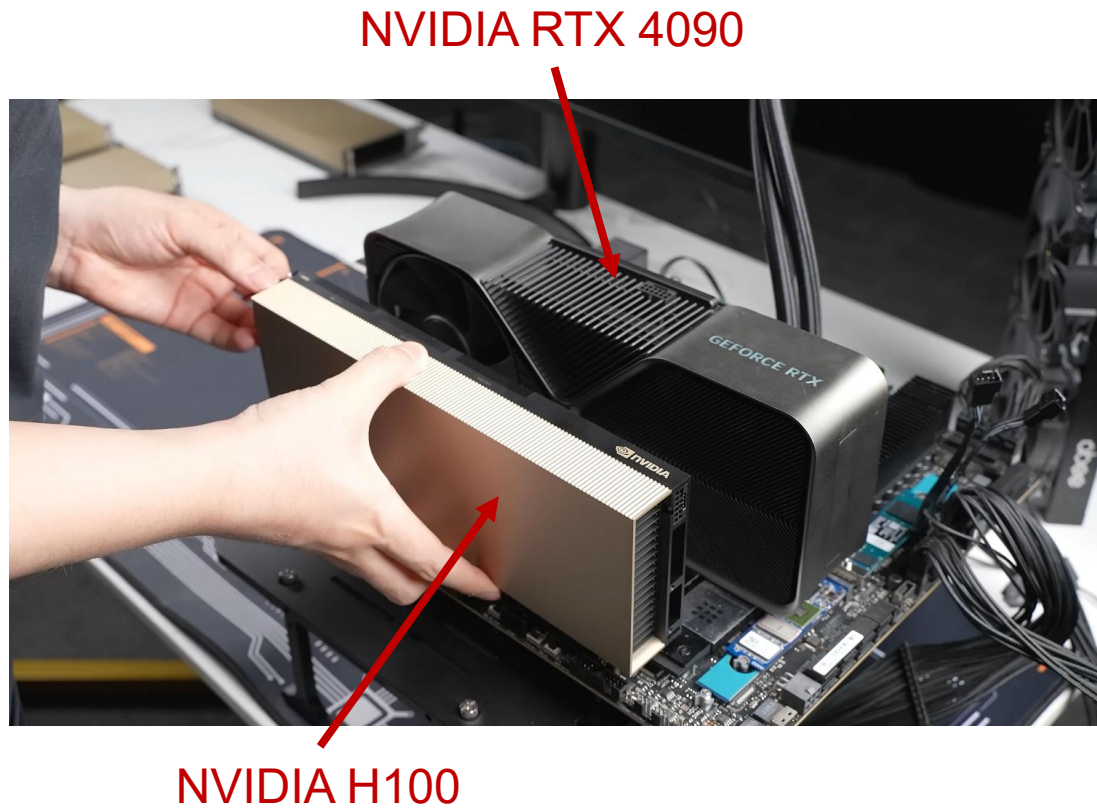


Sizes linear to scale. Selected highlights only. All models are available. All models are Chinchilla-aligned (20:1 tokens:parameters) <https://lilearchitect.ai/chinchilla/> All 300+ models: <https://lilearchitect.ai/models-table/> Alan D. Thompson, 2023-2024.

Source: <https://lilearchitect.ai/models/>

Key question

- Given a **fixed compute budget**, what is the optimal model size and training dataset size for training a Transformer LM?



A Glimpse at NVIDIA B200



Let's say you can use one GPU for one day

- ❑ Would you train a 5 million parameter LM on 100 books?
- ❑ What about a 500 million parameter LM on one book?
- ❑ Or a 100k parameter LM on 5k books?

Observations from Kaplan et al., 2020

- ❑ Performance depends strongly on scale (model params, data size, and compute used for training), weakly on model shape (e.g., depth, width)
- ❑ Perf vs. scale can be modeled with power laws
- ❑ Perf improves most if model size and dataset size are scaled up together. Increasing one while keeping the other fixed leads to diminishing returns
- ❑ Larger models are more sample efficient than smaller models, take fewer steps / data points to reach same loss

Source: Kaplan et al., “Scaling laws for neural language models,” 2020.

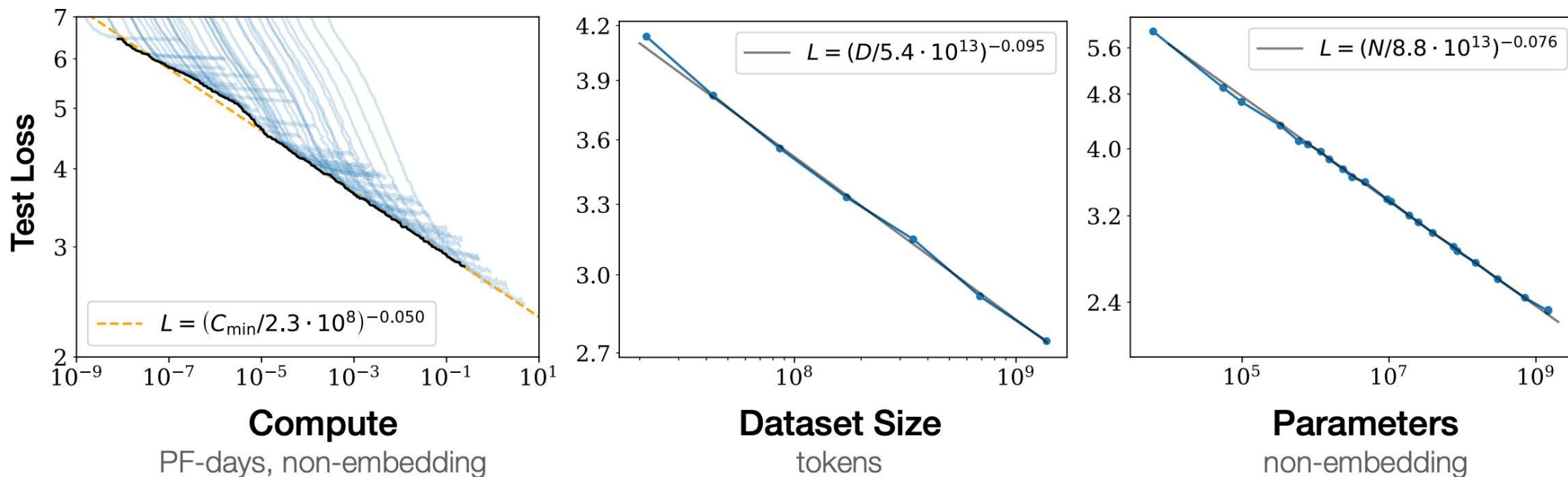
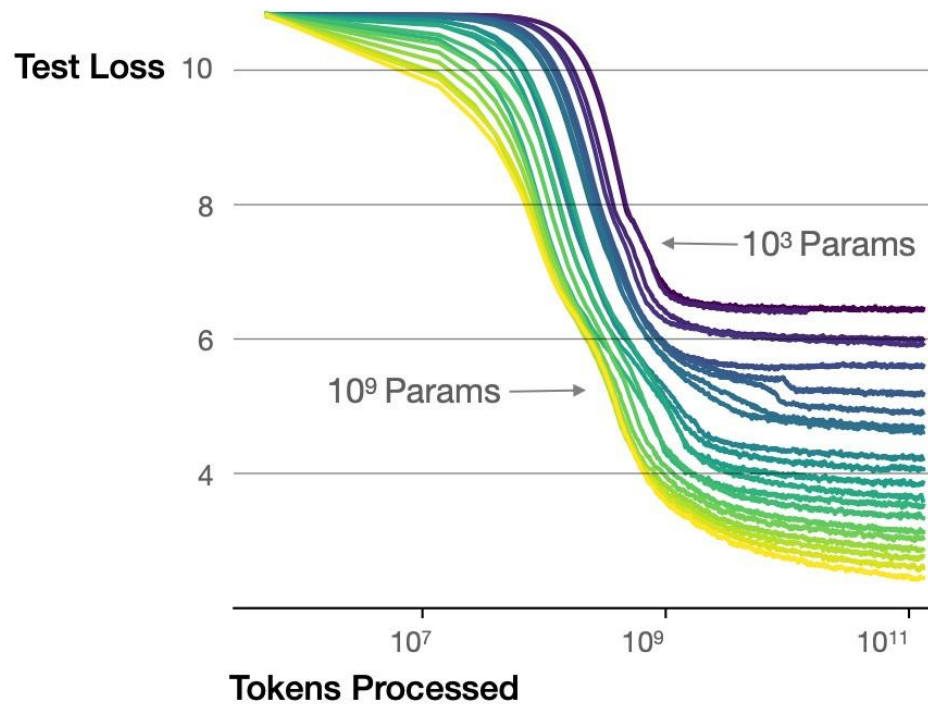


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Source: Kaplan et al., “Scaling laws for neural language models,” 2020.

Larger models require **fewer samples** to reach the same performance



The optimal model size grows smoothly with the loss target and compute budget

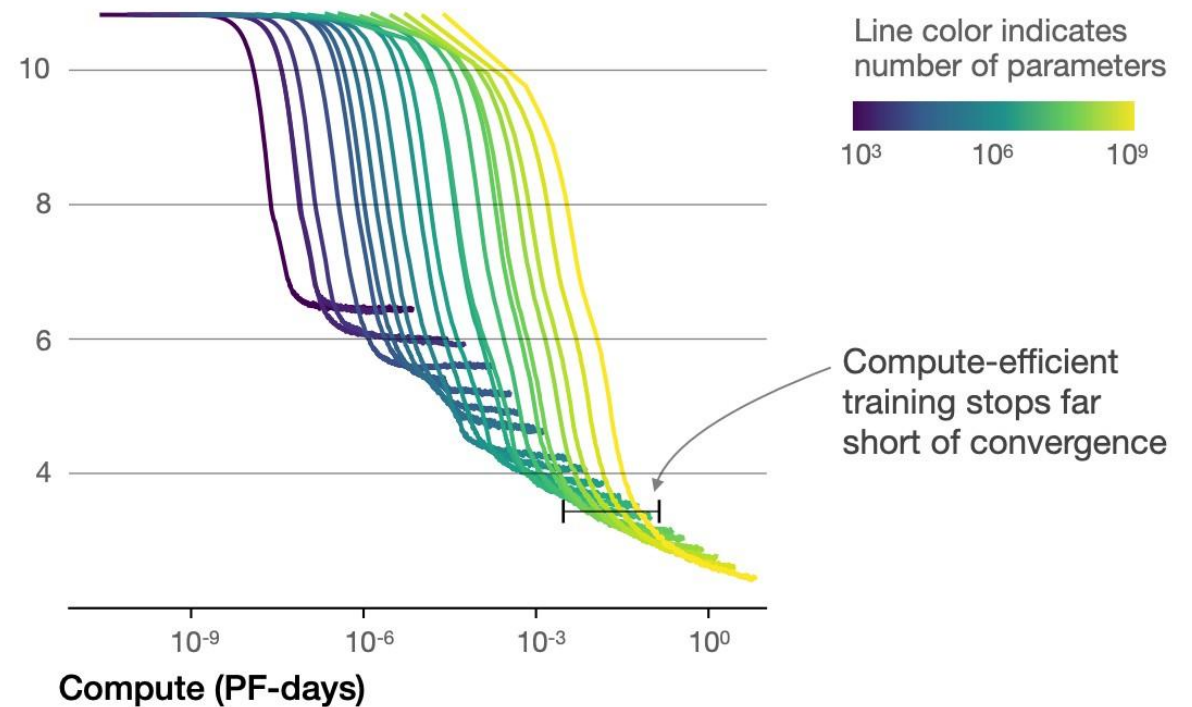


Figure 2 We show a series of language model training runs, with models ranging in size from 10^3 to 10^9 parameters (excluding embeddings).

Source: Kaplan et al., "Scaling laws for neural language models," 2020.

Issues with Kaplan Law

- ❑ Used same learning rate schedule for all training runs, regardless of how many training tokens / batches!
- ❑ This schedule needs to be adjusted based on the number of training steps; otherwise, it can impair performance
- ❑ The resulting “scaling laws” from Kaplan et al., are flawed because of this!

Source: Kaplan et al., “Scaling laws for neural language models,” 2020.

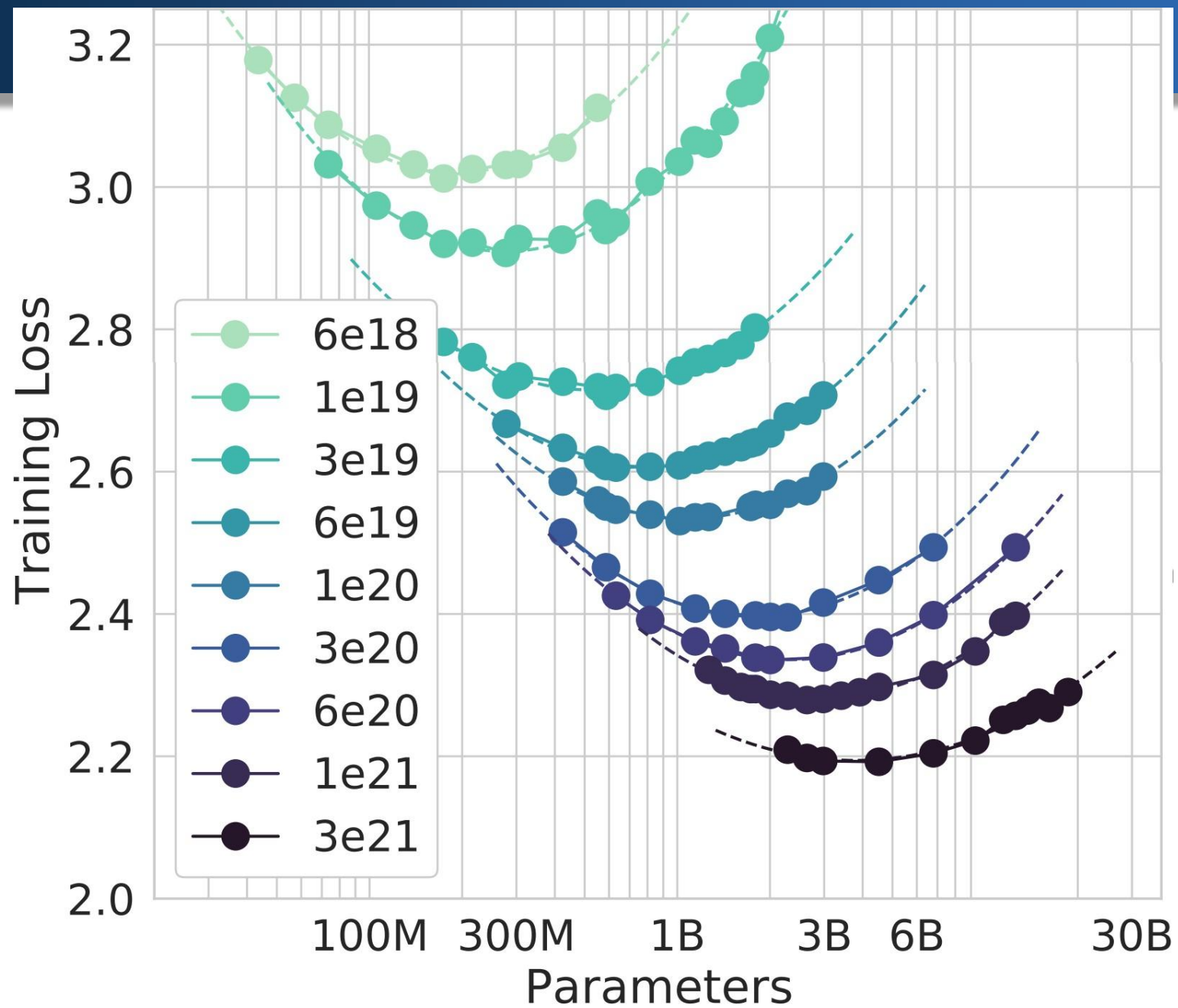


Chinchilla

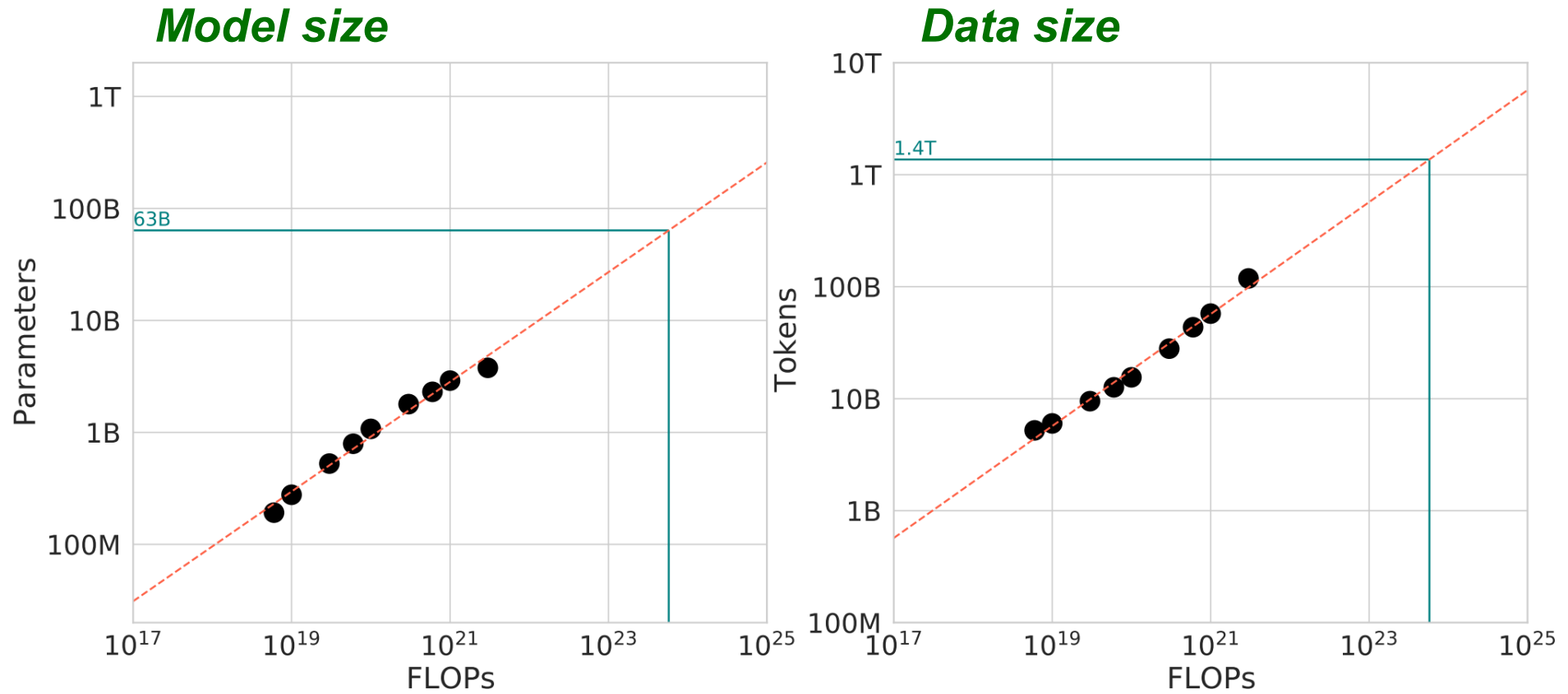
Hoffmann et al., “Training Compute-Optimal Large Language Models,” NeurIPS, 2022.

Quick takeaways

- ❑ Kaplan et al., 2020: if you're able to increase your compute budget, you should prioritize increasing model size over data size
 - With a 10x compute increase, you should increase model size by 5x and data size by 2x
 - With a 100x compute increase, model size 25x and data 4x
- ❑ Hoffmann et al., 2022: you should increase model and data size approximately at the same rate
 - With a 10x compute increase, you should increase both model size and data size by 3.1x
 - With a 100x compute increase, both model and data size 10x



Data and Model Sizes



Chinchilla Scaling Law

- **L** : LM test loss (average cross-entropy loss)
- **D** : dataset size (# of training tokens seen)
- **N** : # of model parameters
- **C** : computer budget (deterministic function of N and D)
 - $FLOPs(N, D)$

- Given a fixed compute budget C

Find $\operatorname{argmin}_{N,D} L(N, D)$

Subject to $FLOPs(N, D)=C$

Chinchilla Scaling Law (cont'd)

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E$$

- E : Test loss achievable by a perfect LM
- $\frac{B}{D^\beta}$: Contribution of the data size
- $\frac{A}{N^\alpha}$: Contribution of the model size
- After training many models, they found the following values:
 - $\alpha = 0.34$, $\beta = 0.28$, $E = 1.69$, $A = 406.4$, $B = 410.7$
- Experiments:
 - Trained two models with the same compute budget C :

Chinchilla Scaling Law: Experiments

□ Gopher (280B params, 300B tokens):

$$\triangleright L(\text{Gopher}) = L(280 \times 10^9, 300 \times 10^9) = \underbrace{0.052}_{\text{finite model}} + \underbrace{0.251}_{\text{finite data}} + \underbrace{1.69}_{\text{irreducible}} = 1.993$$

□ Chinchilla (70B params, 1.4T tokens):

$$\triangleright L(\text{Chinchilla}) = L(70 \times 10^9, 1400 \times 10^9) = \underbrace{0.083}_{\text{finite model}} + \underbrace{0.163}_{\text{finite data}} + \underbrace{1.69}_{\text{irreducible}} = 1.936$$



Gopher



Chinchilla

Source: J. Rae et al., "Scaling language models: Methods, analysis & insights from training gopher." *arXiv preprint arXiv:2112.11446* (2021).

Chinchilla Scaling Law: Recap

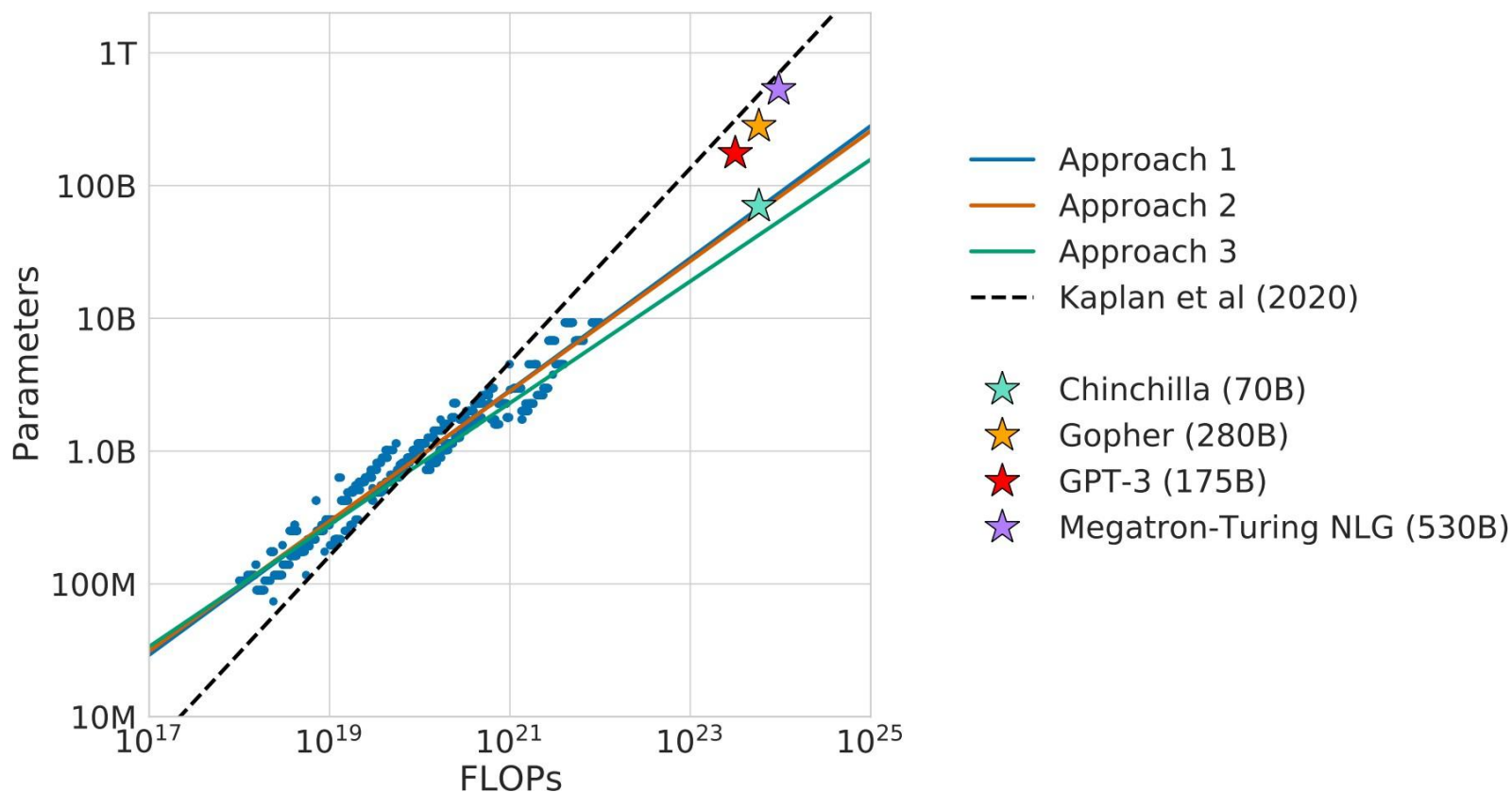
- ❑ You should increase model and data size approximately at the same rate.
- ❑ Chinchilla scaling law tells us that $D \approx 20N$.
- ❑ It is not recommended to train LLMs for less than **200B** tokens though.

Source: J. Hoffmann, et al. "Training compute-optimal large language models." *arXiv preprint arXiv:2203.15556* (2022).

- Given a fixed compute budget, what is the optimal model size and training dataset size for training a Transformer LM?

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
Gopher (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
<i>Chinchilla</i>	70 Billion	1.4 Trillion

- Given a fixed compute budget, what is the optimal model size and training dataset size for training a Transformer LM?



Source: Hoffmann et al., "Training Compute-Optimal Large Language Models," NeurIPS, 2022.

Is Chinchilla model correct?

- Besiroglu et al. reconstructed Hoffmann et al.'s dataset, which resulted in a new fitted model.

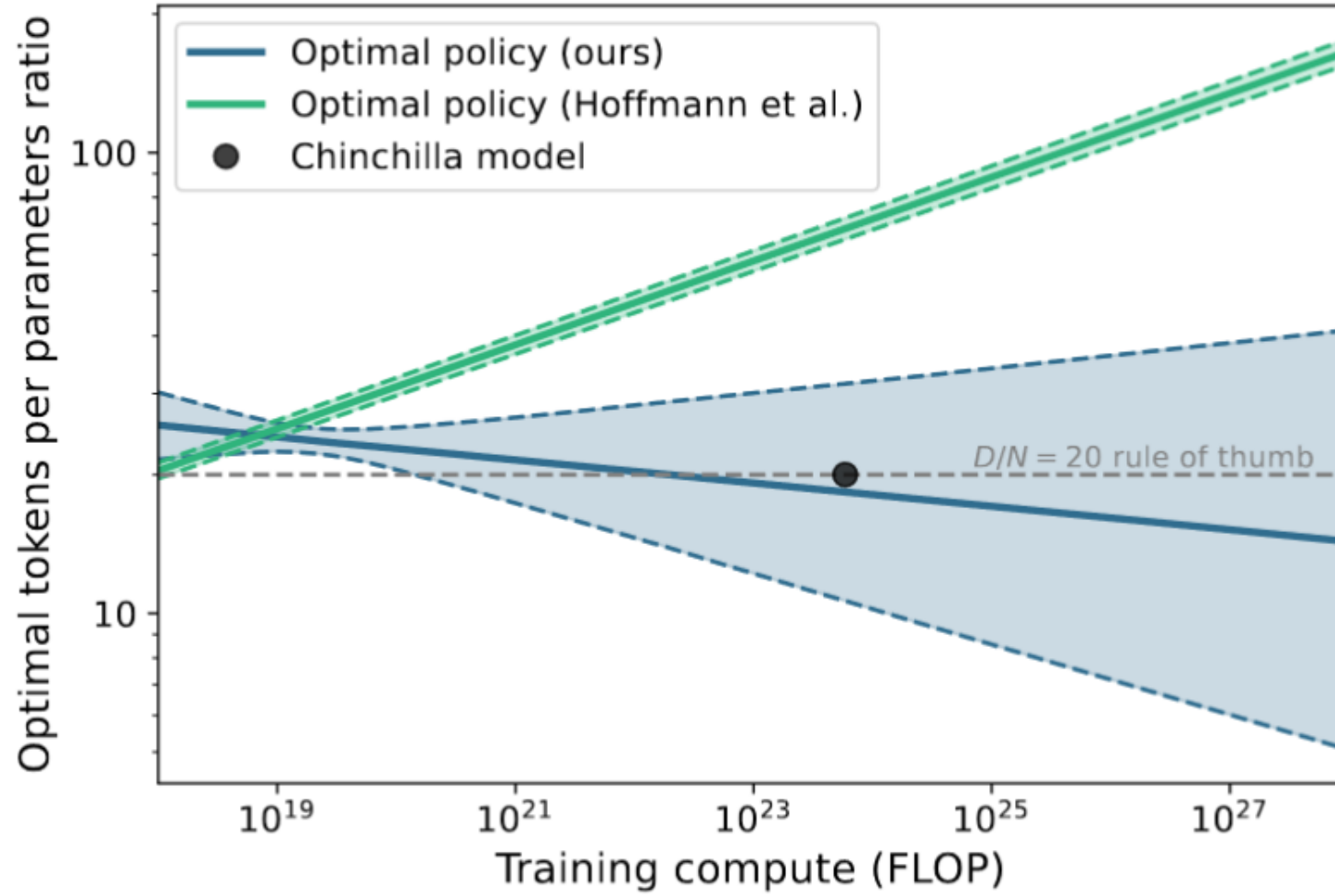
Besiroglu et al.'s model

$$L(N, D) = 1.82 + \frac{514.0}{N^{0.35}} + \frac{2115.2}{D^{0.37}} \quad (3)$$

Hoffmann et al.'s model

$$L(N, D) = 1.69 + \frac{406.4}{N^{0.34}} + \frac{410.7}{D^{0.28}} \quad (4)$$

Source: Tamay Besiroglu et al. "Chinchilla Scaling: A replication attempt." *arXiv preprint arXiv:2404.10102* (2024).



Source: Tamay Besiroglu et al. "Chinchilla Scaling: A replication attempt." *arXiv preprint arXiv:2404.10102* (2024).

Observations

- ❑ The confidence intervals for the optimal scaling policies based on Hoffmann et al.'s estimates are extremely narrow:
 - Intervals this narrow would require **over 600,000 experiments**, while they likely only ran **fewer than 500**.
- ❑ The scaling policy derived from Hoffmann et al.'s estimated parameters suggests using approximately 70 tokens per parameter for optimal performance.
 - This prescription is inconsistent with the 20 tokens-per-parameter ratio actually used by Hoffmann et al. to train their 70B Chinchilla model.



Compute-Optimal Llama Models

Meta's LLaMa 1

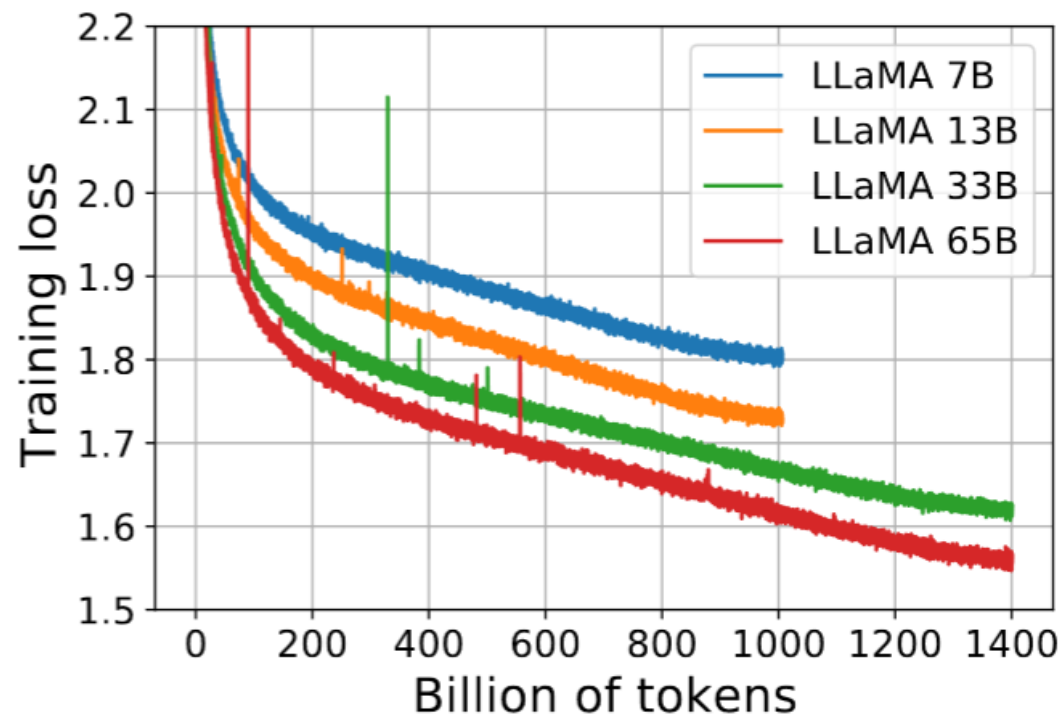


Figure 1: **Training loss over train tokens for the 7B, 13B, 33B, and 65 models.** LLaMA-33B and LLaMA-65B were trained on 1.4T tokens. The smaller models were trained on 1.0T tokens. All models are trained with a batch size of 4M tokens.

Source: H. Touvron et al., "LLaMA: Open and Efficient Foundation Language Models", 2023.

Meta's LLaMa 2

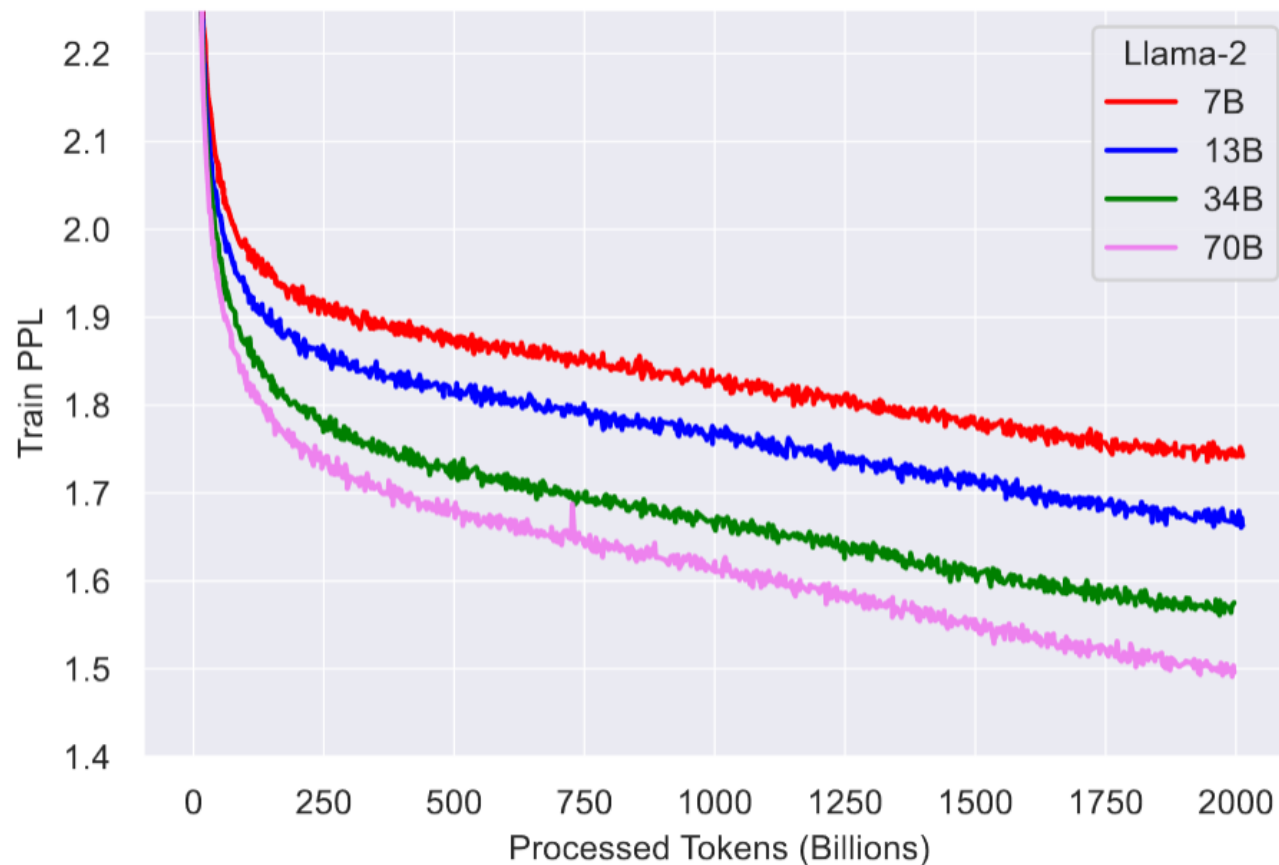


Figure 5: Training Loss for LLAMA 2 models. We compare the training loss of the LLAMA 2 family of models. We observe that after pretraining on 2T Tokens, the models still did not show any sign of saturation.

Source: H. Touvron et al., “LLaMA: Open and Efficient Foundation Language Models”, 2023.

Meta's LLaMa 3

- ❑ The Chinchilla-optimal amount of training compute for an 8B parameter model corresponds to $\sim 160\text{B}$ tokens.
- ❑ Observed model performance continues to improve even after the model is trained on *two orders of magnitude more data*.
- ❑ Both 8B and 70B parameter models continued to improve log-linearly on up to 15T tokens.
- ❑ Larger models can match the performance of these smaller models with less training compute, but **smaller models are generally preferred** because they are *much more efficient during inference*.

Source: <https://ai.meta.com/blog/meta-llama-3/>

Meta's LLaMa 4

- ❑ Llama 4 enables open source fine-tuning efforts by pre-training on 200 languages, including over 100 with over 1 billion tokens each, and overall 10x more multilingual tokens than Llama 3.
- ❑ The overall data mixture for training consisted of more than 30 trillion tokens, which is more than double the Llama 3 pre-training mixture and includes diverse text, image, and video datasets.

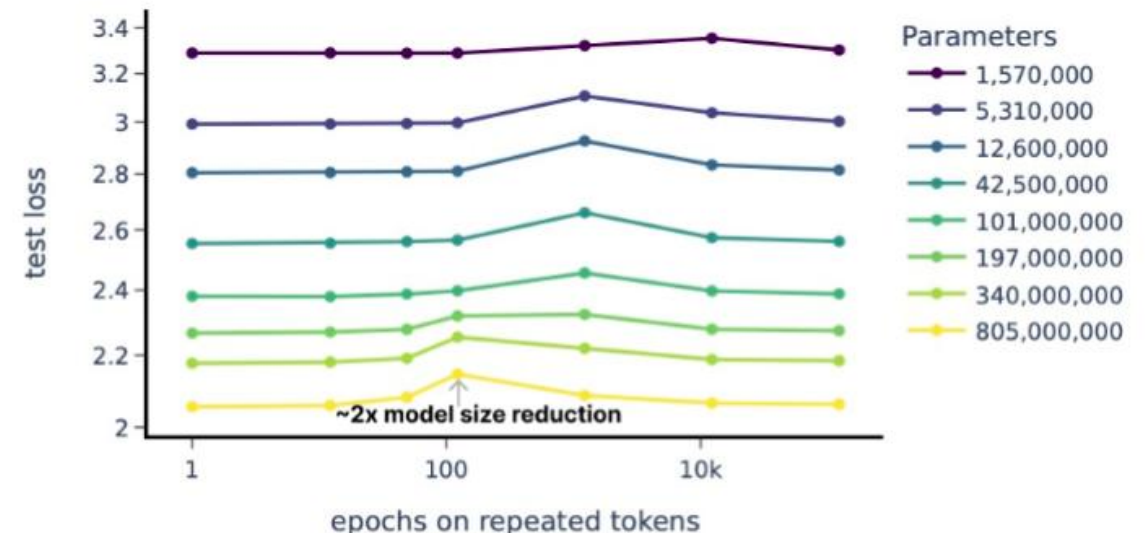
Source: <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>

What about the *type* of data?

What about the type of data?

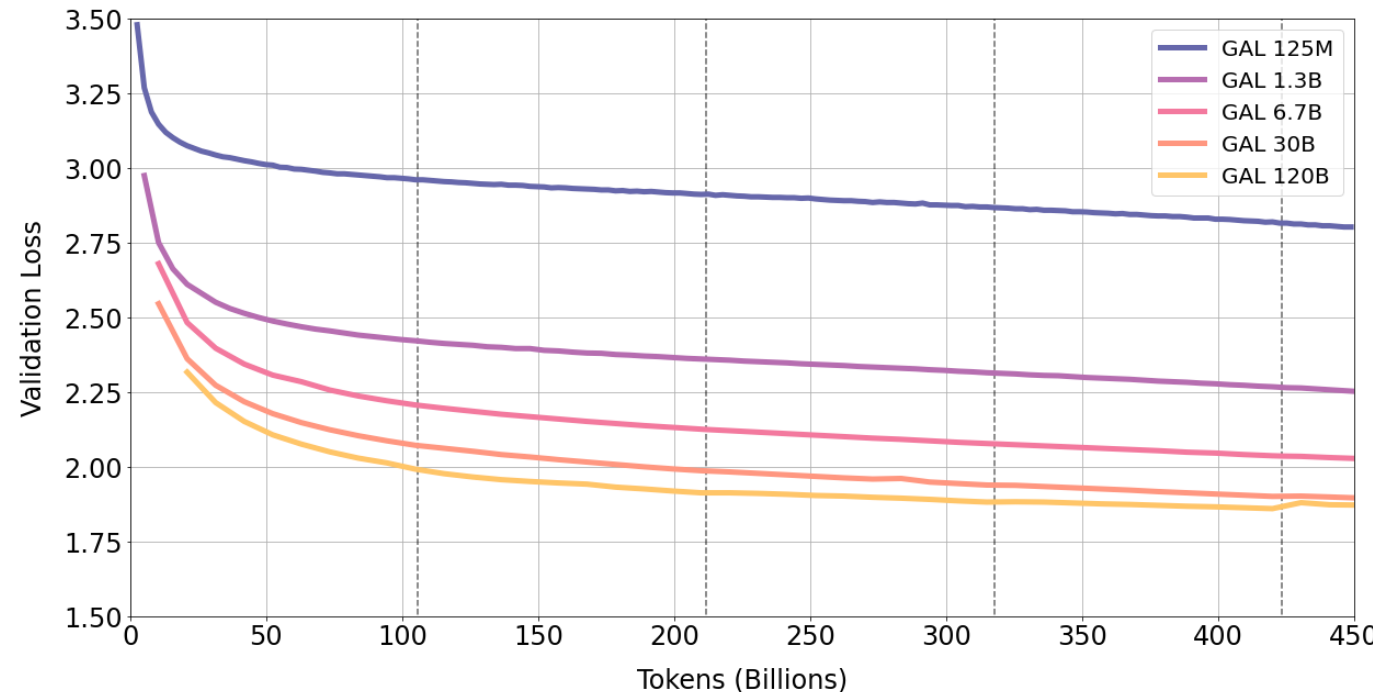
- ❑ The internet contains a huge amount of text, but it's extremely noisy! Copyrighted text (e.g. published books) are much higher-quality, but is it legal to train on them?
- ❑ What is the impact of repeated data?
 - Repeated data can lead to severe degradation in performance (Hernandez et al., May. 2022)
 - “For instance, performance of an 800M parameter model can be degraded to that of a 2x smaller model (400M params) by repeating 0.1% of the data 100 times, despite the other 90% of the training tokens remaining unique.”

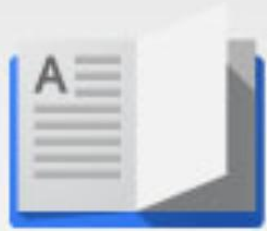
Large Double Descent Effect Caused by Training 10% on Repeated Subset



What about the type of data? (cont'd)

- Repeated data is helpful (Taylor et al., Nov. 2022; Galactica)
 - “We train the models for 450 billion tokens, or approximately 4.25 epochs. We find that performance continues to improve on validation set, in-domain and out-of-domain benchmarks with multiple repeats of the corpus.”
 - “We note the implication that the “tokens $\rightarrow \infty$ ” focus of current LLM projects may be overemphasized versus the importance of filtering the corpus for quality.”





Google Books Search

Books of the world, stand up and be counted! All 129,864,880 of you.

Thursday, August 05, 2010 at 8:26 AM

Posted by Leonid Taycher, software engineer

When you are part of a company that is trying to digitize all the books in the world, the first question you often get is: "Just how many books are out there?"

<http://booksearch.blogspot.com/2010/08/books-of-world-stand-up-and-be-counted.html>

فیلد یو

